

FINANCE 384 Assignment 1

Part A: Return Prediction

Semester Two 2026

Assessment weight	30% of final grade.
Part A	20%
Part B	10%
Part A release date	August 7 th , 2026
Part A due date	September 20 th , 2026 (by 11:59 PM)
Submission method	Canvas

Overview

You are part of a quantitative research team at an equity investment manager. The firm is evaluating whether machine learning should be added to its asset-pricing and stock-selection process.

The current research benchmark is a transparent pooled predictive regression. The chief investment officer wants to know whether one carefully chosen richer model, such as a regularised regression, tree-based method, or neural network, produces better out-of-sample return forecasts and better investment decisions.

Your task is to use the supplied dataset to build, validate, interpret, and govern a return-prediction model. The central question is:

Does a more flexible data-science model improve out-of-sample stock-selection performance enough to justify its complexity, implementation costs, and governance burden?

The chief investment officer is not asking for the most complicated algorithm available. A model earns its place only if it improves a financial decision once benchmark performance, interpretability, and model risk are taken into account. The practical question is whether your team can produce a signal that is strong enough, stable enough, and understandable enough to influence a monthly equity-selection process.

You will work in a group of three (3) to four (4) members.

Data

You will be provided with a dataset. The dataset will contain a panel of stock-month observations. Each row represents one stock at one monthly decision date.

The panel covers S&P 500 index constituents, so the cross-section in any month contains roughly 500 large-capitalisation US stocks. Index membership changes over time, so a stock enters the panel when it joins the index and leaves when it exits.

The Part A release contains:

- `FINANCE384_assignmentA_development_panel.csv`: development observations for model construction, validation, and testing;
- `FINANCE384_stock_month_data_dictionary.csv`: definitions, timing, and roles for the stock-month variables;
- `FINANCE384_market.csv`: monthly equal-weighted excess returns for the S&P 500 constituent universe, provided as `mktrf`.

Use the firm’s fixed chronological split. The split is based on the forecast decision date t :

- **Training sample**: January 1990–December 2014;
- **Validation sample**: January 2015–December 2018;
- **Test sample**: January 2019–November 2022.

This split reflects the firm’s model-validation policy: estimate models on a long historical sample, tune model choices on a later validation period, and reserve a recent period for an untouched pre-implementation backtest. You must not change these periods.

The target variable is the next-month stock excess return:

$$r_{i,t+1}^e,$$

where i indexes stocks and t indexes the month in which the forecast is made. The panel contains `ret_excess_t`, the excess return realised *during* month t .

The feature variables, denoted by $z_{i,t}$, contain information available at or before the forecast date. The data dictionary’s `role` column sorts the non-target columns into three groups:

- **Identifiers** (`date`, `permno`, `ticker`, `comnam`, `gvkey`) label the observation. They must not be used as predictors.
- **Classifications** (`industry`, `ff49_code`, `ff49_name`) record the industry grouping. They are known at month t and *may* be used as predictors if you encode them appropriately. Whether to include them is a modelling choice you should justify.
- **Characteristics** are the numeric predictors: firm fundamentals, lagged returns, risk measures, liquidity measures, and market states. Every one of them is measurable by the end of month t , so all are admissible predictors of $r_{i,t+1}^e$. The `role` column further records which are predetermined at $t - 1$ and which use month- t information.

Feature Engineering and Preprocessing

You are expected to build a reproducible feature matrix rather than passing the raw panel directly into every model. At minimum, your notebook must:

- state your predictor list explicitly, excluding identifiers, realised outcome variables, and anything not available at the forecast date;
- encode categorical variables, such as the industry classifications, with appropriate dummy variables if you choose to use them as predictors;
- define the missing-data rule, scaling rule and any transformations applied before final test evaluation.

Interactions and nonlinear transformations are permitted, but they are not required unless they are part of your chosen richer model. If you include them, explain the economic mechanism they are meant to capture.

Both the pooled OLS benchmark and the richer model must be based on the same base predictor information.

Financial Decision

Treat the model as supporting an equity-selection decision. Each month it predicts next-month excess returns for the stocks in the universe, and the investment team may use those forecasts to form a long-short portfolio or conclude that the model is not strong enough to use.

The required economic evaluation is a monthly, prediction-sorted quintile sort, applied identically to the pooled OLS benchmark and to your richer model so that any difference in performance reflects the forecasts rather than a different trading rule. It produces a bottom portfolio $P1$, a top portfolio $P5$, and the long-short spread $P5 - P1$.

Your analysis should therefore answer:

- a. **Prediction question:** which model forecasts returns most effectively out of sample?
- b. **Investment question:** do the forecasts improve a realistic stock-selection or portfolio decision?

Required Model-Fitting Protocol

Use the following protocol for the assessed comparison.¹

1. Fit the benchmark pooled OLS model on the combined training and validation samples.
2. Fit the richer model specification using the training sample only, and select the hyperparameters using the validation sample only. The primary validation criterion is the pooled validation MSE (or, equivalently, the pooled validation RMSE), with lower values preferred. Once selected, retain the training-sample fitted model without re-estimating it on the combined training and validation samples.

¹Rolling-window and expanding-window estimation schemes are not permitted.

3. Report validation metrics for the selected richer model. Evaluate the pooled OLS benchmark and the selected richer model on the test sample. Validation metrics are not required for the pooled OLS benchmark.

For the assessed submission, present a comparison of exactly one richer model with the pooled OLS benchmark. Your selection should be justified.

Required Metric Definitions

For the test comparison, compute metrics for the pooled OLS benchmark and the richer model using the same stock-month rows.

Pooled RMSE. Compute RMSE across all valid stock-month rows in the relevant period:

$$\text{RMSE} = \sqrt{\frac{1}{\sum_t N_t} \sum_t \sum_{i=1}^{N_t} \left(r_{i,t+1}^e - \hat{r}_{i,t+1}^e \right)^2},$$

where N_t is the number of valid stocks in month t .

Rank correlation. For each month, compute the Spearman rank correlation between predicted and realised next-month excess returns. Use average ranks for ties.

Portfolio construction. For each valid month and each model:

- sort stocks by predicted next-month excess return from low to high;
- assign stock k out of N_t sorted stocks to quintile $q_{k,t} = 1 + \lfloor 5(k-1)/N_t \rfloor$, so $P1$ contains the lowest-ranked stocks and $P5$ the highest-ranked stocks;
- calculate each portfolio's realised return as the equal-weight average of the constructed target $r_{i,t+1}^e$;
- calculate $P5 - P1$ as the monthly $P5$ return minus the monthly $P1$ return.

Report the average monthly excess return for $P5 - P1$, with its t -statistic, for the richer model in the validation and test periods and for the pooled OLS benchmark in the test period.

Market-model alpha. Estimate the monthly market-model alpha for $P5 - P1$ using:

$$r_{t+1}^{p,e} = \alpha_p + \beta_p \text{MKT}_{t+1} + u_{t+1}^p,$$

where $r_{t+1}^{p,e}$ is the $P5 - P1$ portfolio excess return and MKT_{t+1} is the `mktrf` market excess return from the supplied file. The `date` variable in the market file identifies the month in which `mktrf` is realised, so a portfolio formed at date t must be matched to `mktrf` dated $t+1$. Report monthly α_p and its t -statistic for the richer model in the validation and test periods and for the pooled OLS benchmark in the test period.

Documented Analysis Plan

Your notebook must contain a short design statement with:

- the benchmark model and the richer model to be compared;
- the hyperparameter grid or search rule used for tuning;
- the feature-engineering and preprocessing choices.

Part A: Model Development and Investment Recommendation (20%)

Tasks

Task A.1: Frame the modelling problem.

- Define the target variable ($r_{i,t+1}^e$ from `ret_excess_t`), feature set, and prediction timing.
- Define the feature-engineering and preprocessing steps used to build the model.
- Apply the firm-specified train, validation, and test periods.

Task A.2: Estimate a transparent benchmark model.

Estimate a pooled predictive regression of the form:

$$r_{i,t+1}^e = \alpha + \beta' z_{i,t} + \varepsilon_{i,t+1}.$$

This model is the benchmark against which the richer model is judged. Use the same base predictor information as the richer model.

Task A.3: Estimate one richer prediction model.

Select exactly one richer model from the Finance 384 toolset for the assessed comparison with the pooled OLS benchmark. Acceptable choices include:

- ridge regression;
- lasso regression;
- elastic net;
- regression tree;
- random forest;
- gradient-boosting model;
- a feed-forward neural network implemented with TensorFlow/Keras.

Your model choice should be justified from a financial economics perspective. Do not use a complex model only because it is available. Explain what modelling feature it adds relative to pooled OLS, such as shrinkage, variable selection, nonlinearities, interactions, or flexible function approximation, and why that feature could plausibly matter for expected-return prediction.

Task A.4: Estimate and tune the richer model using the training and validation data.

- Hyperparameters must be selected on the validation dataset (i.e., without using the test period).
- Once the hyperparameters have been selected, retain the model fitted on the training sample. Do not re-estimate it on the combined training and validation samples.
- Your notebook must make the tuning process reproducible.
- Your tuning grid should be large enough to be meaningful but small enough to run in the specified execution environment in reasonable time.

Task A.5: Evaluate out-of-sample prediction performance.

Report pooled RMSE and mean monthly Spearman rank correlation for the richer model in the validation and test periods. Compare the pooled OLS benchmark and the richer model on both metrics in the test period. Report the number of months and stock-month rows behind each figure.

Task A.6: Evaluate economic performance.

Translate the predictions into the required quintile portfolios. Report the average monthly $P5-P1$ excess return and its t -statistic, together with the market-model alpha and its t -statistic, for the richer model in the validation and test periods and for the pooled OLS benchmark in the test period.

Task A.7: Interpret the model.

- Explain the trade-off between predictive performance and interpretability for your chosen model.
- Discuss the differences between the richer model's validation and test metrics in terms of accuracy and portfolio performance.
- Identify any other points of interest in your model and its performance.

Task A.8: Make an investment recommendation.

Recommend one of the following:

- advance the richer model to paper trading or implementation testing;
- retain the pooled OLS benchmark;
- delay implementation pending additional validation.

Your recommendation must be supported by the evidence in your report and notebook.

Part A Deliverables

One group member should submit Part A via Canvas. The submission must contain the following files:

1. A **model discussion report** in PDF format.

- Maximum length: two pages for the main discussion, excluding references.
- Include an additional mandatory one-page technical appendix defining the model, hyperparameters, feature-engineering choices, and preprocessing steps.
- An additional results appendix of up to two pages (containing additional validation and test tables or figures) is optional.
- The report should explain the benchmark model, the richer model, validation results, test evidence, and recommendation.
- The report and appendix should define modelling choices and summarise evidence in prose, tables, or figures. Code excerpts, notebook-cell references, and implementation details belong in the submitted notebook.
- The intended audience is the chief investment officer and the quantitative research manager.

2. **Executed analysis notebook**, named `FINANCE384_GroupXX_Analysis.ipynb`.

- Submit the notebook with all cells run in order and all output visible.
- The notebook must reproduce every table and figure used in the report.
- Use clear Markdown headings and comments.
- Include a documented analysis plan table.

3. **Team process and contribution statement**.

- Briefly state each group member's contribution and any unresolved contribution issues.

References

Your submission should cite any external sources used to motivate methods, or interpret asset-pricing variables. Use APA style. References do not count toward the report limits.

Grading Rubric

Part A: Model Development and Investment Recommendation (20 marks)

Marks are awarded separately for technical execution and financial justification. The descriptors indicate the overall standard expected.

Criterion	Marks	A range	B range	C range	D or fail
Technical	8	6.5–8 Accurate, complete, and reproducible implementation of the required protocol. Modelling, evaluation, portfolio construction, and outputs show strong technical control; any issues are minor.	5.5–6 Sound and mostly complete implementation. Some minor omissions or inconsistencies are present, but the main results remain credible and reproducible.	4–5 A recognisable implementation with several gaps, weak choices, or departures from the required protocol. The results are usable but require caution.	0–3.5 Major errors, missing components, leakage, or reproducibility problems materially weaken the analysis.
Financial justification	12	9.5–12 Financially persuasive model choice and interpretation. Prediction and portfolio evidence, validation and test differences, complexity, and implementation are weighed with judgement. The recommendation is clear and well supported for the intended audience.	7.5–9 Plausible financial motivation and generally sound interpretation. The evidence supports the recommendation, although some trade-offs or implications receive limited attention.	6–7 Some relevant financial reasoning is present, but the links among model choice, evidence, and recommendation are uneven, generic, or only partly developed.	0–5.5 Financial justification is weak, largely descriptive, or disconnected from the results. The recommendation is absent, unclear, or poorly supported.